

Lab Activity – Lecture 5: Transforming Data into Information (Advanced)

Objective

Students will work with a real dataset, apply formulas, perform data analysis, create charts, and apply professional formatting.

Time Plan (60 Minutes)

- Dataset Entry: 10 mins
- Formula Application: 20 mins
- Problem Solving: 15 mins
- Charts Creation: 10 mins
- Formatting: 5 mins

Part 1: Dataset (Student Evaluation System)

Name	Attendance (%)	PF	ICT	Math	Total	Average	Grade	Status
Ali	85	70	80	75				
Sara	92	85	90	88				
Ahmed	60	60	65	70				
Fatima	95	95	92	89				
Bilal	45	40	55	60				

Part 2: Formulas

Total: =SUM(C2:E2)

Average: =AVERAGE(C2:E2)

Grade: =IF(G2>=85,"A",IF(G2>=70,"B",IF(G2>=60,"C","Fail")))

Status: =IF(OR(B2<75,C2<50,D2<50,E2<50),"Fail","Pass")

Part 3: Problem-Based Questions

1. Who is the Topper?
2. How many students failed due to attendance?
3. How many students failed overall?
4. What is the class average?

Part 4: Charts (Step-by-Step)

Bar Chart (Student Performance):

1. Select Name and Total columns.
2. Go to Insert → Column/Bar Chart.
3. Choose Clustered Column.
4. Add chart title: "Student Performance".

Pie Chart (Pass vs Fail):

1. Count number of Pass and Fail students.
2. Create a small table (Pass | Fail).
3. Select the table.
4. Go to Insert → Pie Chart.
5. Add chart title: "Result Distribution".

Part 5: Professional Formatting

- Highlight Fail students in Red.
- Highlight Pass students in Green.
- Bold headings.
- Apply borders to table.
- Adjust column width.
- Center align all data.